

Survival Model Evaluation: `chicago_licences`

Fitted with the strategy-survival-model pipeline; methodology validated separately against synthetic ground truth

Source data	<code>datasets/chicago_licences.csv.gz</code>
Rows	262,763 (83.6% with the ending observed, 16.4% censored)
Start dates	2002-01-02 to 2026-07-31
Evaluation	5 expanding-window temporal folds
Source of figures	<code>reports/chicago_demo/metrics.json</code> , regenerated by the command in section 6

1. Summary

This report evaluates a survival model fitted to `chicago_licences.csv.gz`, 262,763 rows, each observed from its start date for 'licensed_days' days with 'closed' marking whether the ending was seen (83.6%) or the row was still running when observation stopped (16.4%). Median observed duration is 759 days. The model predicts, from the 300 features available at the start date, how long each row survives.

Pooled over 157,658 out-of-time test rows, the XGBoost AFT model reaches a concordance index of 0.697 (95% bootstrap interval 0.695 to 0.699), against 0.500 for a coin flip. Set beside a Cox proportional hazards baseline on the same features it scores 0.695 to the baseline's 0.695; both of those are fold means, which is the only like-for-like basis and is why the pooled figure is not the one compared. At the 1095-day horizon its censoring-weighted Brier score is 0.206, beating the 0.247 of a no-skill forecast that assigns every row the population average.

That bootstrap interval is narrow, and it is worth saying what it does and does not cover. It holds each fold's fitted model fixed and resamples the test rows, so it measures how precisely this run's score is pinned down by the number of rows scored, nothing more. It says nothing about how much the score would move on a different stretch of history, and the per-fold spread of 0.604 to 0.782 in Table 2 is the better guide to that. The rows are also not independent, since licences in the same trade and the same year fail together, which makes the true interval wider than the one printed.

This dataset has no known generating process. Unlike the synthetic validation report, there is no oracle ceiling to say how much signal remains unclaimed, and feature attributions cannot be checked against a true mechanism. What carries over from the synthetic validation is the pipeline itself: the same temporal folds, label re-censoring, likelihood-based selection, and calibration checks, verified there against known answers.

2. Method

Rows are ordered by start date and evaluated with 5 expanding-window folds; every fold trains only on rows that started before its split date and tests on the next block, so training sets grow from 105,103 to 231,220 rows. Training labels are re-censored at each split date. An ending recorded after the split is rewritten to "still running at the split", which is all a model trained then could have known. Test labels keep their full outcomes.

The model is XGBoost with the `survival:aft` objective; censoring enters through interval labels. Hyperparameters are selected on the first fold's training window by held-out censored log-likelihood (selected depth 3, loss scale 0.9), and the predictive log-normal scale is calibrated separately on a temporal tail slice (final value 1.28). Numeric features pass through as-is, missing values included, which XGBoost handles natively; the Cox baseline receives train-window median imputation. Text columns are one-hot encoded.

3. Results

Table 1. Concordance index, pooled across folds. Higher is better; 0.500 is a coin flip.

METHOD	HARRELL C	95% INTERVAL
XGBoost AFT (pooled)	0.697	0.695 to 0.699
XGBoost AFT (fold mean)	0.695	not resampled
Cox proportional hazards (fold mean)	0.695	not resampled

On this dataset the Cox baseline outscores the boosted model, 0.695 against 0.695 on fold-mean concordance. Both fitted models are saved and the Cox model is the one this run recommends for scoring new rows. Reporting that, rather than presenting the boosted model as the result, is the point of carrying a baseline at all. Each fold refits the Cox model, and its risk scores are relative ones centred on that fold's own training window, so they are meaningful within a fold but carry no common scale across folds; the fold-mean rows are the like-for-like comparison and the pooled figure is not comparable to them.

Table 2. Per-fold results. Censoring is the share of each fold's test rows whose ending was not observed.

FOLD	SPLIT DATE	TRAIN N	TEST N	CENSORED	AFT	COX
1	2009-11-09	105,103	31,532	6%	0.701	0.695
2	2012-08-02	136,592	31,532	14%	0.604	0.617
3	2014-06-09	168,159	31,532	16%	0.707	0.706
4	2017-10-23	199,693	31,531	27%	0.680	0.672
5	2022-02-23	231,220	31,531	60%	0.782	0.787

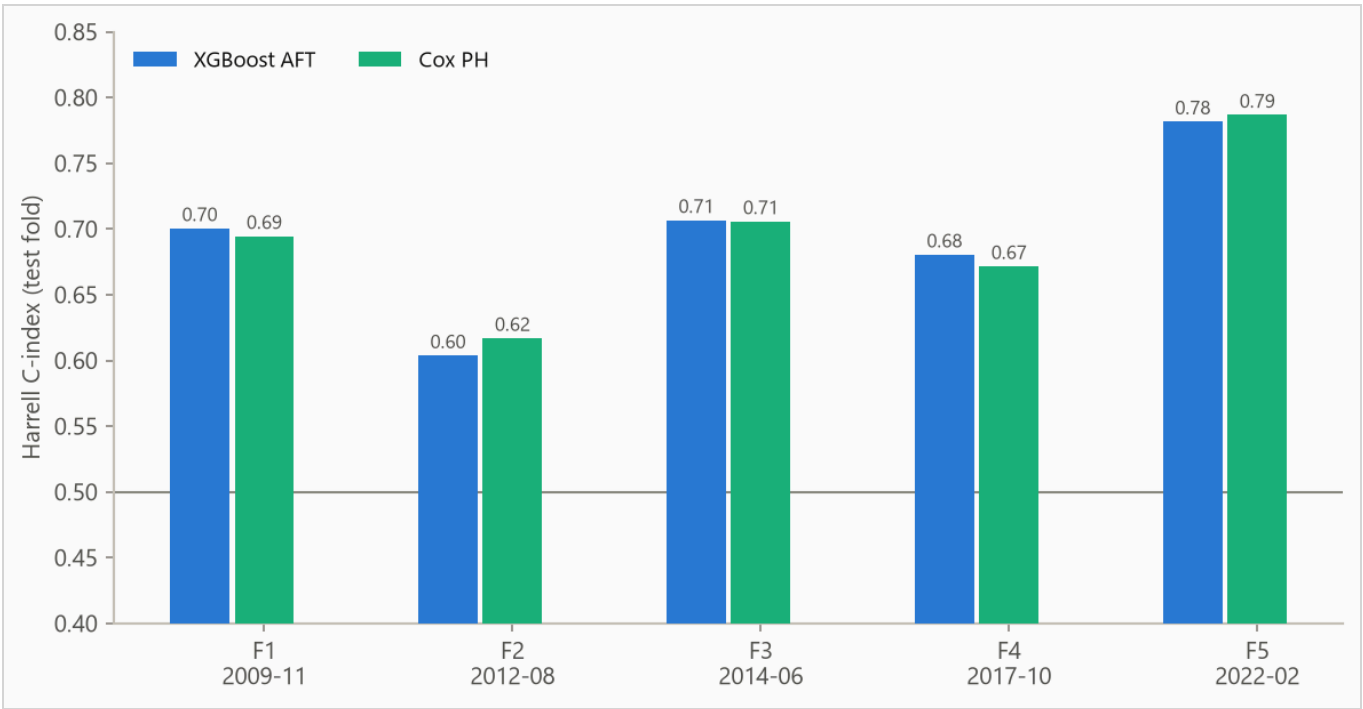


Figure 1. Concordance by fold, from 0.604 to 0.782. Folds differ in censoring mix, so cross-fold comparisons are indicative rather than exact.

Table 3. IPCW Brier score by horizon; lower is better. The no-skill column assigns every row the pooled Kaplan-Meier marginal.

HORIZON	AFT	COX	NO-SKILL MARGINAL
365 days	0.084	0.082	0.158
1095 days	0.206	0.207	0.247
1825 days	0.194	0.192	0.215

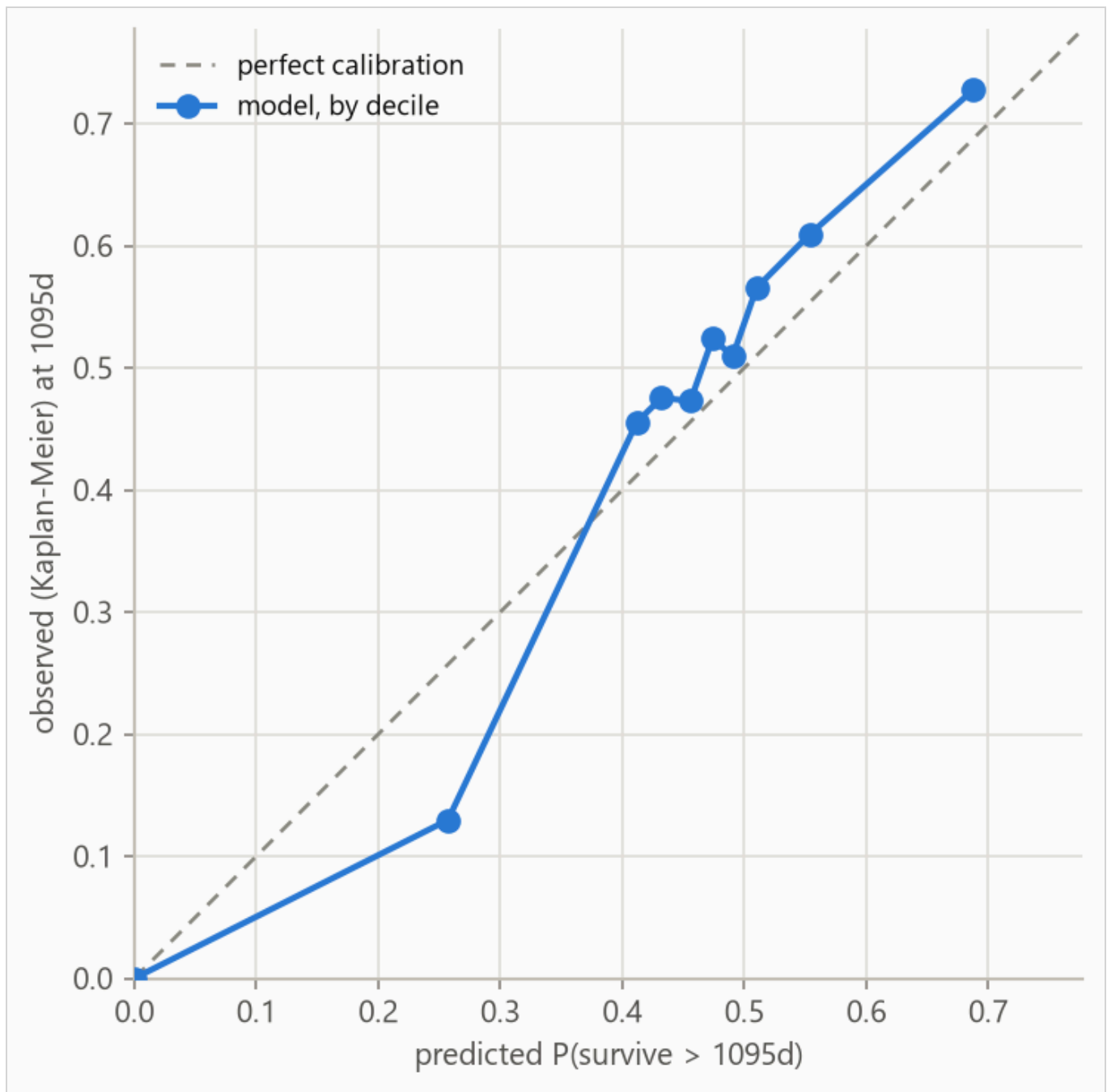


Figure 2. Predicted against observed survival at 1095 days, by predicted decile. Observed frequencies are Kaplan-Meier estimates within each bin, so censored rows contribute correctly.

Table 4. Decile calibration at 1095 days, the values plotted in Figure 2.

DECILE	N	PREDICTED	OBSERVED (KM)	DEVIATION
1	15797	0.000	0.000	-0.000
2	15748	0.257	0.130	-0.127
3	16166	0.412	0.455	+0.044
4	15480	0.432	0.475	+0.043
5	17372	0.456	0.473	+0.017
6	14036	0.474	0.524	+0.050
7	15761	0.490	0.509	+0.019
8	16054	0.510	0.566	+0.055
9	15601	0.553	0.609	+0.056
10	15643	0.688	0.728	+0.040

4. What the model uses

Feature attributions are computed on the log-time margin, where positive values push predicted survival up. On this data they are descriptive only. There is no known mechanism to check them against, and correlated features can split credit in ways that reflect the model's internal choices as much as the data, so read the ranking as "what this model leaned on", not as importance in the world.

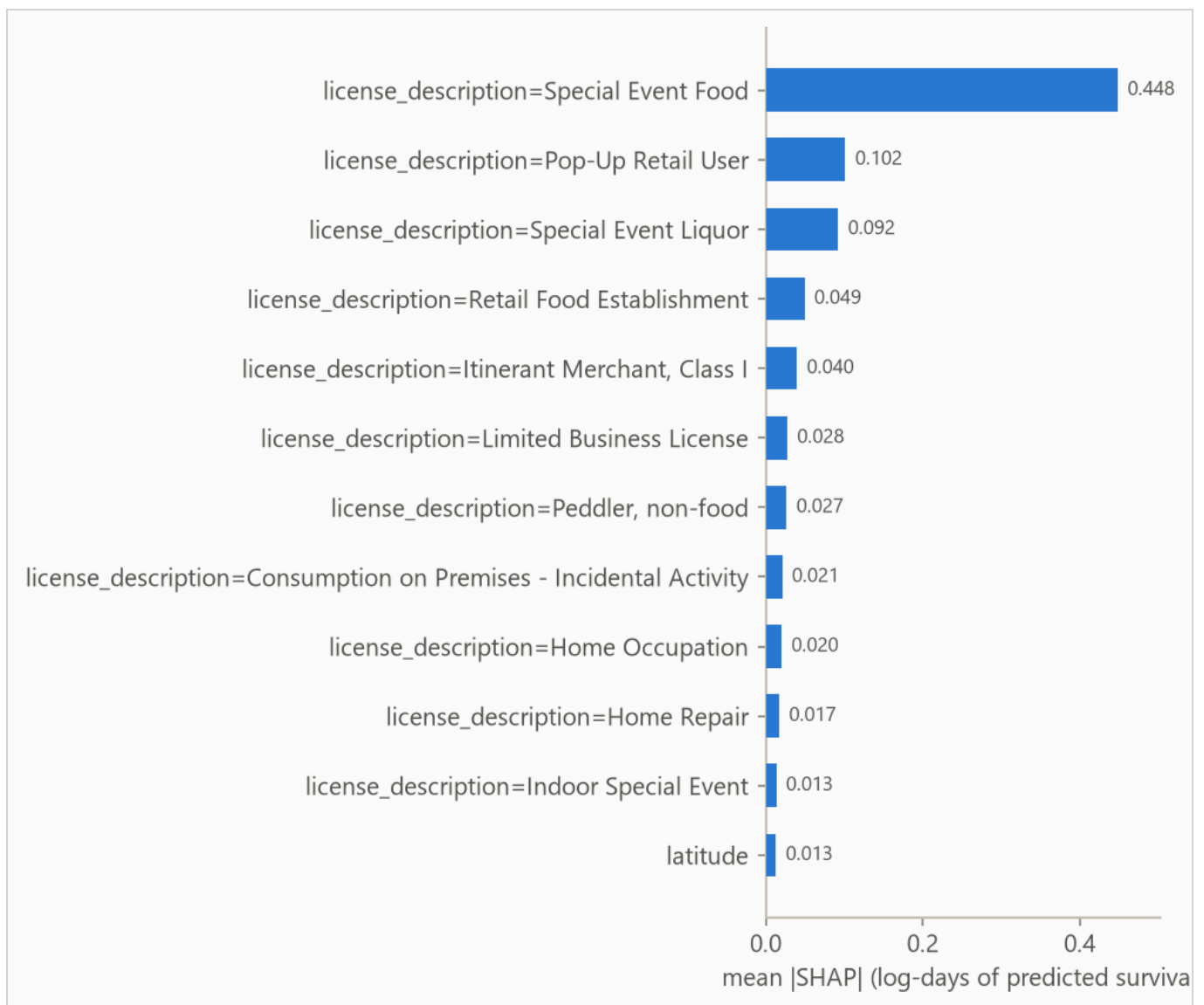


Figure 3. Mean absolute attribution by feature.

Table 5. Top eight features by mean absolute attribution.

FEATURE	MEAN ATTRIBUTION
license_description=Special Event Food	0.448
license_description=Pop-Up Retail User	0.102
license_description=Special Event Liquor	0.092
license_description=Retail Food Establishment	0.049
license_description=Itinerant Merchant, Class I	0.040
license_description=Limited Business License	0.028
license_description=Peddler, non-food	0.027
license_description=Consumption on Premises - Incidental Activity	0.021

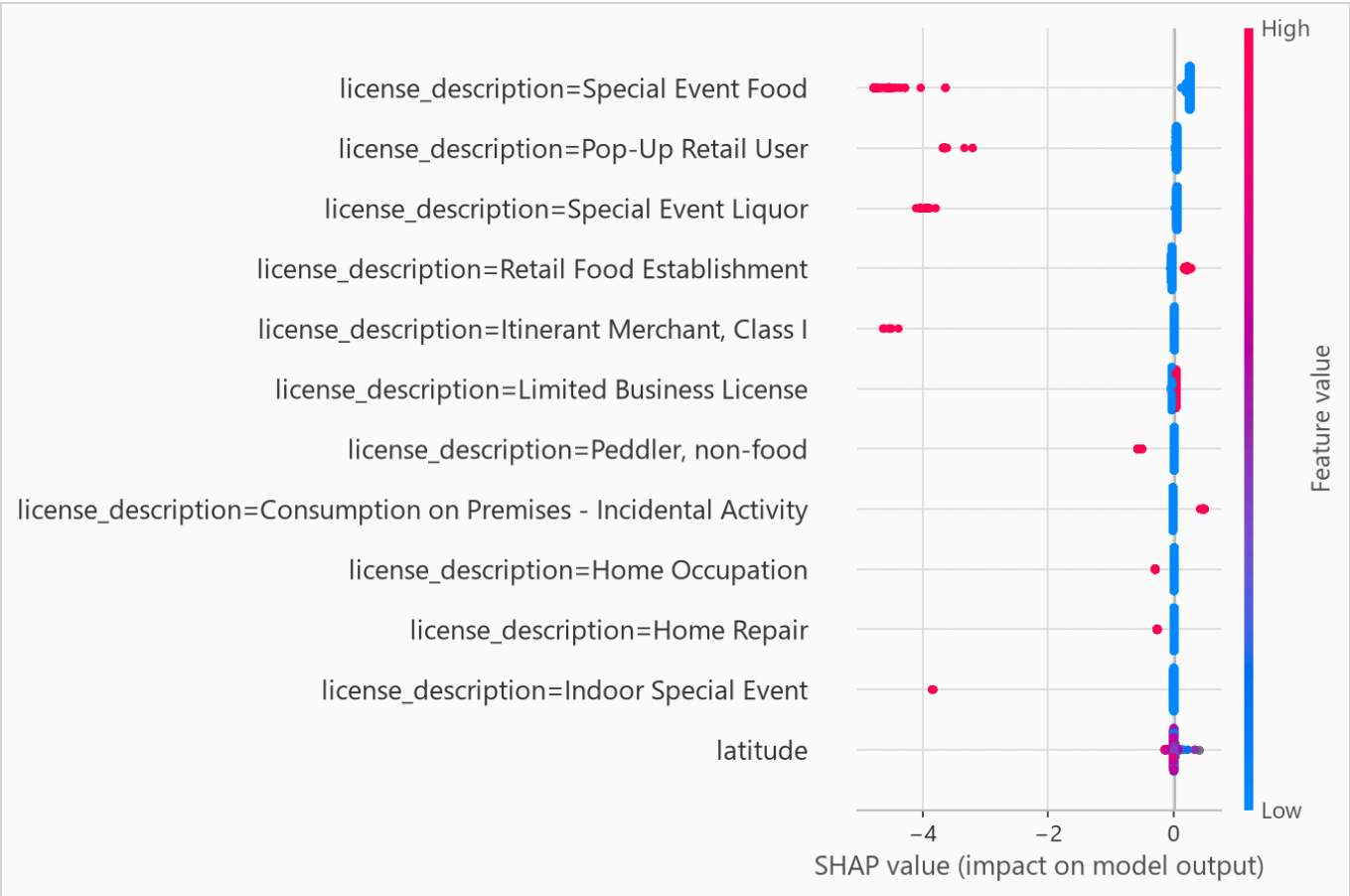


Figure 4. Per-row attributions across the explanation sample.

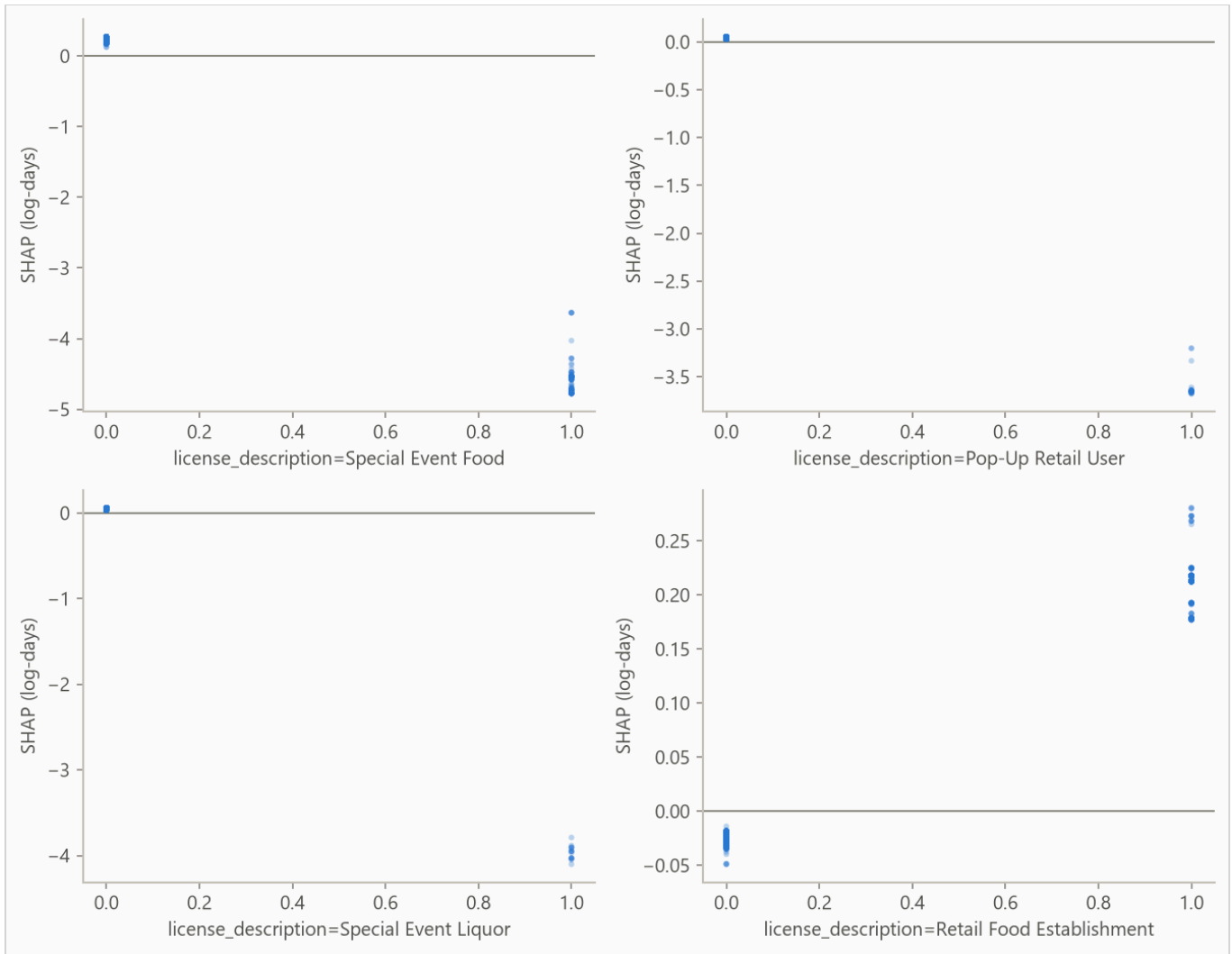


Figure 5. Attribution against feature value for the four features with the largest mean attribution.

5. Limitations

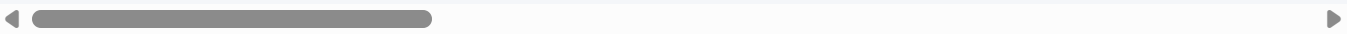
No ground truth. Nothing bounds how much predictable signal this dataset contains, so the concordance above cannot be placed relative to a ceiling the way the synthetic validation could.

Censoring may be informative. The evaluation assumes rows stop being observed for reasons unrelated to their risk. If observation ended early on rows that were about to end anyway, survival estimates are biased upward. Whether that holds here depends on how this dataset was collected.

Harrell's concordance is biased under heavy censoring. Fold-to-fold comparisons in Table 2 should be read with each fold's censoring share in mind.

6. Reproducing this run

```
python scripts/run_fit_evaluate.py --data datasets/chicago_licences.csv.gz --name chicago_licences
python scripts/run_build_report.py --run reports/chicago_demo
```



Generated from `reports/chicago_demo/metrics.json` by `scripts/run_build_report.py --run`. 262,763 rows,
157,658 out-of-time test rows.